WORLD POPULATION TRENDS AND CO_2 EMISSIONS

PROJECT REPORT

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1 Abstract

This study examines world population trends and their relationship with CO2 emissions using population and CO2 emissions datasets. Exploratory data analysis was conducted to examine the structure, completeness, duplicates and outliers in the population dataset. Population projections for 2030 were calculated using the exponential growth model. Data visualization was used to identify 10 most populous countries and examine their CO2 emission trends. Correlation analysis was performed to investigate the relationship between 2022 population size and 2019 CO2 emissions per capita. The correlation coefficient was 0.00817, indicating an almost nonexistent linear relationship. Furthermore, continental comparisons showed that Asia had the highest sum of country-level per-capita emissions (242), while South America had the lowest (29.4). The outcomes showed that population size alone does not explain differences in CO2 emissions per capita.

2 Introduction

Population growth and CO2 emissions are important factors in understanding global environmental challenges. The increase in population may influence energy consumption, industrial activities and high demand for natural resources. However, countries with larger populations do not necessarily have higher CO2 emissions per person. This project analyzes population trends and CO2 emission patterns using data management, exploratory data analysis, visualization and statistical techniques in R.

3 Methods and Dataset Description

Two datasets were used: the World Population Dataset, containing 234 observations and 17 variables, and the CO2 Emissions Around the World dataset, containing 215 observations and 35 variables. The population dataset included country names, population figures for different years, area, density, growth rate and continent. The CO2 dataset contained annual CO2emissions per capita from 1990 to 2019. R programming was used for importing both datasets,examining them, data cleaning, manipulation and visualization. Exploratory data analysis(EDA) involved examining variable names, data types, head/tail datasets, duplicates, missing values and outliers. The exponential population projection formula was applied to estimate population in 2030. The dplyr, tidyr and ggplot2 packages were used for data manipulation and visualization. Correlation and scatter plots were used to examine the relationship between population and CO2 emissions.

4 Results and Findings

The exploratory analysis showed that the population dataset contained 234 countries and 17 variables, with no missing values or duplicated observations. However, several quantitative variables contained outliers, particularly population and area.

Population projections for 2030 showed differences in expected growth across countries. For example, Afghanistan was projected to reach approximately 50.52 million people, while Algeria was projected to reach approximately 51.20 million people.

In 2022, China was the most populous country with 1.426 billion people, followed by India with 1.417 billion. The United States ranked third with approximately 338.29 million people. Population trend visualizations showed changes in population among the ten selected countries between 1990 and 2022.

Correlation analysis revealed a moderate positive relationship between land area and world population percentage (r=0.4533). Other relationships were weak. The correlation between 2022 population size and 2019 CO2 emissions per capita was approximately r=0.00817, indicating almost no linear association.

The continental comparison showed that Asia had the highest sum of country-level per-capita emission values (242), followed by Europe (229). North America ranked third (79.6), while South America recorded the lowest value (29.4). These values represent sums of per-capita emissions, not actual continental total emissions.

5 Discussion

The findings indicate that population growth varies across countries, with China and India having the largest populations in 2022. The population projections of 2030 demonstrate how differences in growth rates can influence future population sizes.

The correlation coefficient of 0.00817 suggests that population size has almost no linear relationship with CO2 emissions per capita in the merged dataset (merged_data). This is indicated by energy consumption, industrial activities and economic development among countries.

The comparison among continents showed that Asia had the highest sum of country-level per-capita emissions, followed by Europe. However, these sums should not be interpreted as total continental emissions because the analysis used per-capita values rather than total emissions.

6 Lessons Learned and Recommendations

This project improved my level of understanding more about data imprtation, data management, exploratory data analysis, data visualization and correlation analysis using R. I became more familiar with using R programming com-

mands,learnt how mathematical models can be applied to project future population sizes.

For population trends, learnt the relation between World population and CO2 emissions,the impact CO2 emissions have on the environment . Additional variables, such as GDP, industrialization and energy consumption.Furthermore, population projections should be interpreted cautiously because growth rates may change over time. And here are some recommended links that were used to help in analysing our worls Population and CO2 trends //https://www.kaggle.com/code/devraai/world-gdp- population-co2-analysis? https://https://www.kaggle.com/datasets/iamsouravbanerjee /world-population-dataset?

7 Conclusion

This study examined world population trends and their relationship with CO2 emissions using statistical analysis and visualization in R. The results identified China and India as the most populous countries in 2022 and highlighted differences in population growth across countries. The correlation analysis revealed almost no linear relationship between population size and per-capita CO2 emissions. Overall, the study demonstrates the importance of combining population, economic and environmental indicators to better understand global emission patterns.